Lab session 2-3

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1 Motivation reduction - How it works

In these two lab sessions I was asket to apply reduction to the summation of a matrix. Reduction is a methodology that avoids race conditions and still allows threads to execute together. Another solution is to use atomicAdd(), this function ensures that only 1 thread can access the memory at a time, this method is of course slow and not recommended.

the principle of reduction is to do the summation in several steps. The first step sums the first with the second / third with the fourth / fifth with the sixth (/ ...) element in, for example, an array. These sums are then placed at position 0, 2, 4 in the array. The next step is to sum these 'sums' in the same way as before, this is repeated until everything has been added to the first element. It is noted that here, the threads do not write memory to the same location at the same time (no race conditions).

It is also important that the threads should be synchronized at every 'step'. If this does not happen, a thread could get to the next 'step' faster and still write memory together with another thread (race condition).

2 Timing results

2.1 array

The following table shows that the GPU time increases as the width increases. The CPU also increases, but is still much faster. We can conclude that the summation of an array is not enough computation to draw conclusions. It is noticeable that the GPU time increases more from 512. This is because after this value (256 threads) more threads are used than can be executed in parallel by the video card.

width	time CPU (us)	time GPU (us)
8	1	112
16	0	116
32	1	119
64	1	124
128	2	131
256	2	148
512	4	154
1024	8	221
2048	10	327

2.2 matrix

To obtain more computing power, a matrix can be used. Summing a matrix with reduction requires more computing power. As can be seen from the following table, the time of the GPU is still much longer than that of the CPU. It can be noted that the GPU time between a height of 300 and 500 increases less percentageally than the CPU time.

height (width = 2048)	time CPU (us)	time GPU (us)
100	587	5273
300	1856	18529
500	3291	24359